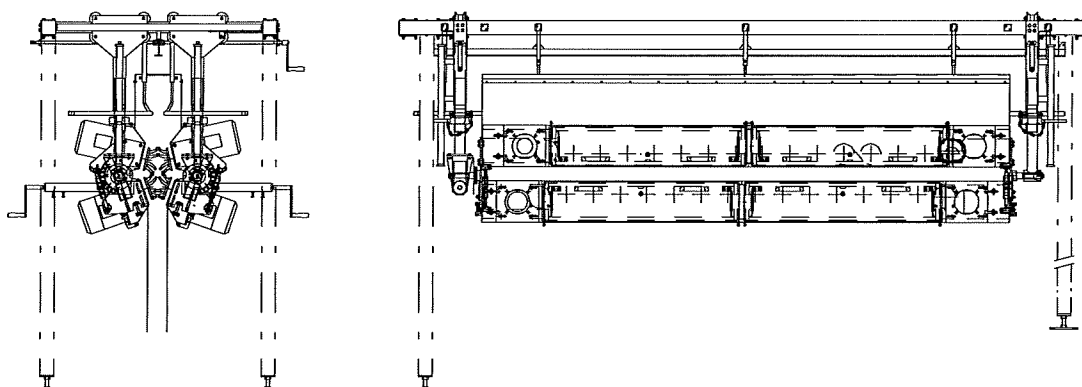


User- and Maintenance manual



Manual: 2005090066 (732447) Version: 1 GB

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Foreword

Meyn Food Processing Technology B.V. is one of the worlds leading designers and manufactures of systems and equipment for the poultry processing industry. Meyn's flexible service and support organisation and its constant emphasis on product innovation through extensive research and development, enable Meyn to supply today's customers with the solutions of tomorrow.

Regardless of the size or complexity of the process, Meyn will supply its customers with situation specific solutions that are critical in reaching their objectives. Therefore, Meyn combines a solid knowledge of the customers business with expert process design skills. This is crucial, as only well focused solutions generate the desired returns.

Our ambition is to supply solutions that support fully controlled in line processing; handling and tracking every product or batch throughout the entire facility while controlling the quality of the output, combined with the ability to match the incoming birds with the sales orders in the most optimal manner.

Meyn is worldwide represented by a network of subsidiaries and agents. In order to supply complete integrated solutions to our customers, Meyn has a number of strategic partners in the following processing sectors:

- Bonescan.
- Mechanical deboning.
- Logistic systems.
- Refrigeration.
- Water treatment.
- Vacuum transport.
- Rendering.
- Convenience food.

This manual

This manual is intended for the operator(s) of this machine. It's a means for information and instruction, so you can use the machine correctly and safely.

The manual consists of 7 chapters and a few appendices:

- Chapter:1. Introduction, includes general information about the machine and the supplier.
- Chapter:2. Safety, includes a description of the most important safety aspects associated with the use of the machine.
- Chapter:3. Installation and set-up, includes a description of the actions that must be performed before the machine can be used.
- Chapter:4. Operation, the actions that must be performed during normal operation.
- Chapter:5. Cleaning, includes a description of the cleaning activities that must be carried out to keep the machine in good condition and to facilitate safe and hygienic operation.
- Chapter:6. Maintenance, includes a description of the maintenance activities that are necessary to keep the machine in good condition and to facilitate safe operation.
- Chapter:7. Disassembly/assembly and adjustments describes the disassembly and assembly of parts needed by reinstallation of the machine to another place or environment and to ensure that the machine performs safely and efficiently.
- Appendix:. Appendices, includes aspects/details for the machine.

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1 Introduction

This chapter contains technical data about the machine and gives a description of the goal and working of the machine.

1.1 Technical specifications

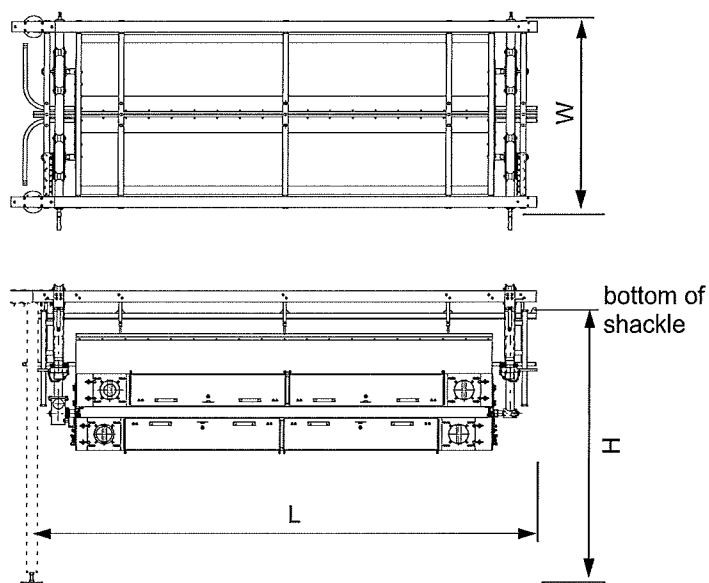


Figure 1.1 Plucker JMD2

<i>Specifications:</i>			
Capacity	8,000 BPH	Weight	1,900 kg
Live bird weight	1000-3500 g.	Noise level	less than 70 dB(A).
Shackle pitch	6 inch.	Water consumption	1.8 m3/h
Number of Plucking disc	64	Water connection	1" ext. (2x)
Number of plucking fingers	640	Number of motors	8
Length (L)	5,010 mm	Motor type	Emod TM132M6A
Width (W)	1,600-2,400 mm	Motor capacity	8x 3 kW
Height (H) b.o.s.	1,800-2,000 mm		

Table 1.1 Specifications

1.2 General arrangement drawing

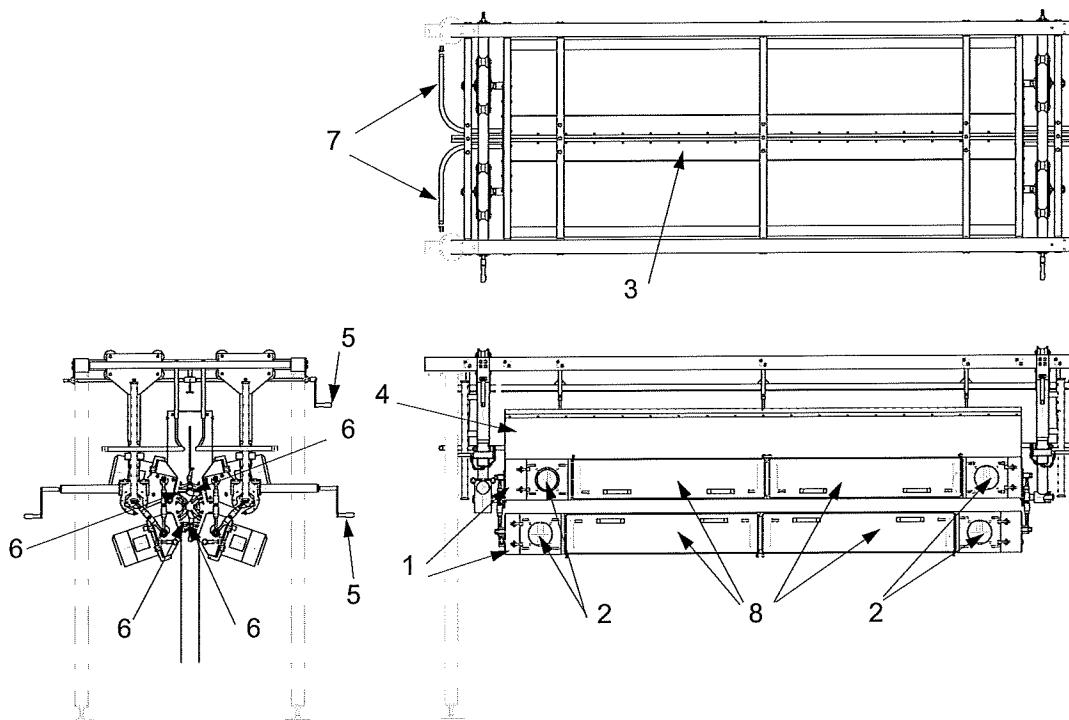


Figure 1.2 Plucker JMD 2

The plucker consists of the following components:

1. Plucker beams (4x).
2. Electric motor (8x).
3. Feather roof.
4. Curtain.
5. Crank handle.
6. Plucking disks and fingers
7. Water pipelines.
8. Doors.

1.3 What is a Meyn Plucker JMD2

The plucker is designed to form part of the killing line. A killing line consists of various machines, each of which plays a part in the killing process. The birds are transported through the machines by means of an overhead conveyor. Each machine operates independently and contributes to the delivery of a clean product, prepared for the evisceration and further processing.

The purpose of the plucker is to remove the feathers from the chickens. When the carcass is heated in an earlier process (hard scalding), the epidermis is also removed from the birds.

Feathers are attached to the skin by means of hair follicles. Soaking the bird in hot water destroys the attachment. This process is referred to as scalding. Following this process, the feathers can be easily removed by rubbing the skin with the hands or with mechanical rubber fingers.

Pluckers are mounted in a plucking line, which consists of different types of pluckers, depending on the capacity of the line and the type of chicken. This ensures an optimum result.

The chickens, hanging vertically by the legs on the shackles of the overhead conveyor, are guided through the pluckers. The pulleys and plucking disks are driven in opposite directions by motors and drive belts. This process, enhanced by the speed at which it occurs, ensures that the feathers are 'twisted' out of the birds.

Water pipes are used for the continuous removal of feathers. Loose feathers are kept wet, so that they fall onto the transport belt or into the collecting bin. The wet feathers are pumped away, washed and subjected to further treatment.

1.4 The end product

The plucker is designed to achieve the following:

- To remove all the feathers from the chicken.
- To remove the skin when scalding is used (hard scalding).
- To ensure that the skin of the bird is not damaged.
- To ensure that no part of the bird is broken or damaged.



Check the output of the machine regularly.
If the output is unsatisfactory, please read chapter 7.
Disassembly/assembly and adjustments.

1.5 The Meyn Plucker technical

The plucker is constructed from stainless steel and synthetic materials. The plucking disks are manufactured from synthetic material or from stainless steel.

The plucker consists of the following main components:

- Four plucker beams.
- Eight electric motors.
- 64 plucking discs.

The plucker is fitted with 64 rotating disks, each having 10 rubber fingers. The disks are driven by eight electric motors and are positioned such that the shape of the chicken can be accurately followed. During the plucking process, a water sprayer flushes the plucked feathers away.

The plucker is adjustable in height, width and at an angle. The correct adjustment are dependant on the size of the birds, the scalding temperature etc. For more information see chapter 7.

2 Safety

This chapter contains a description of safety aspects relating to the machine. It is particularly important that you are fully aware of the contents of this section. The Meyn Food Processing Technology B.V. has taken all possible steps to provide you with all the necessary information concerning any dangers that may exist when the machine is used. The customer is responsible for understanding and applying these safety rules.



If any of these aspects are not clear, please ask the manufacturer for an explanation.

2.1 The most important safety risks

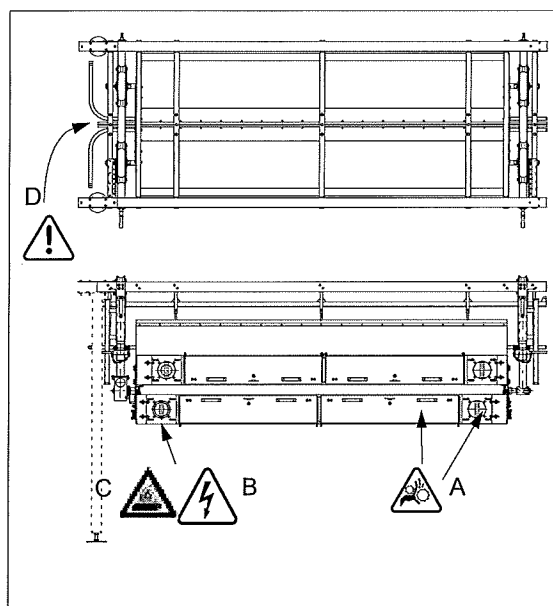


Figure 2.1 Safety risks.

- A hand or an article of clothing can become trapped in the drive mechanism between the motors and the pulleys (A).
- With abnormal operation, the electric motors may cause electric shocks (B).
- During production, the electric motor housing become hot, which can result in burns (C).
- An operator can become trapped in the rotating parts within the plucker (D).

2.2 Safety devices

The following safety features are incorporated to make this machine as safe as possible to use:

- The machine is constructed in such a way that the plucking beams can be easily pushed outwards. The advantage of this is that the inside of the plucker is easily accessible for maintenance and repairs and is easy to clean. This greatly reduces the probability of exposure to rotating parts and, with normal operation, safety is guaranteed.
- The drive mechanism between the motors and the pulleys is protected by an encasing which can be closed by doors. Do not open the doors during operation!
- During production the motor heats up. We advise you not to touch the motor during production and for some time afterwards.
- The electric motors are protected by an encasing.
- You will find information and instructions on how to maintain and use the drive motors safely in Appendix C.

2.3 Safety measures to be taken

Before and during use of the machine it is important to take a number of safety measures and instructions into account, which are described in the sections below.

2.3.1 Preparing for safe operation

For safe operation of the machine and preceeding any act, it is essential that a number of safety measures will be taken: these are mentioned in the following paragraph.

- First read the concerning instructions given in this manual, before you perform any work on the machine.
- Wear footwear which prevents slipping; mind loose-hanging clothing.
- You must take note of all the safety symbols attached to the machine.
- Check that the power supply complies with the locally applicable regulations.
- Ensure that the machine is correctly placed in a stable position.
- Ensure that the operation, minor maintenance activities and activities concerned with installing or moving the machine are only performed by qualified personnel, that is to say by personnel who have the necessary skills and are familiar with the contents of this manual.
- Modifying the machine in any way, could endanger the safety and proper functioning of the machine.

2.3.2 Safety measures during operation

For safe operation of the machine, it is essential that a number of safety instructions are taken during an act: these are mentioned in the following paragraph.

- Ensure that no persons are on, in or under the machine while it is in use.
- Keep the area around the machine free from materials and other obstacles.
- Always stop the machine if a product becomes jammed.
- Never put your hands or any other part of your body close to rotating parts while the machine is running.
- Mind to place warning notices with the control panel of the machine while performing maintenance operations to the machine.

2.4 Know the pictograms

Machines manufactured by Meyn Food Processing Technology B.V. are provided with safety and protective devices. Even so, it is important to be careful when you are performing any operation on the machine. The following pictograms indicate possible dangers. The pictograms are applied to the machine and are also printed in this handbook at those places where risky activities are described.



Check regularly to ensure that all the pictograms are still in the correct positions on the machine. If any pictogram has become detached or unreadable, apply a new one immediately.



Danger
High voltage



Danger
Moving parts



Warning
Hot surface



Danger
Warning

3 Installation

This chapter gives a description of the actions that must be performed before the machine can be used.

3.1 Installing the Plucker

The plucker is supplied as a fully assembled machine. It is delivered with all adjustable parts of the machine already adjusted. The machine only needs to be placed in the plucking street of the slaughtering line and connected to the power supply.

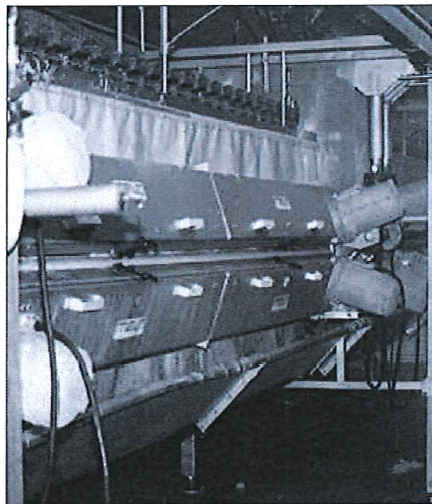


Figure 3.1 The plucker in the plucking line.

The positioning of the plucker in relation to the overhead conveyor is important for an efficient operation. The centre axis of the plucker must coincide with the centre line of the overhead conveyor.

The distance between the scalding and the plucking line must be kept as short as possible to ensure that the chickens cool off as little as possible. Hot water can be sprayed into the first plucker to improve the result.

Check all adjustments after installation. Section 7, 'Disassembly/assembly and adjustments', describes the adjustments that must be maintained for optimum machine performance.

3.1.1 Required floor area

Floor area required: The surface area occupied by the machine is given in section 1.1, Technical specification, (L*W). Nevertheless, it is desirable to install the machine on a floor area that makes it easy to clean and maintain it. Meyn advises you to keep a clear space of 1 m on all sides of the machine.

3.2 The connection of the Plucker

After installing the plucker, the electric motors must be connected to the power supply, the water pipes must be connected to the water supply and the water drainage must be connected.



Before connecting the machine, check to see if the voltage and frequency are the same as those stated on the type plate

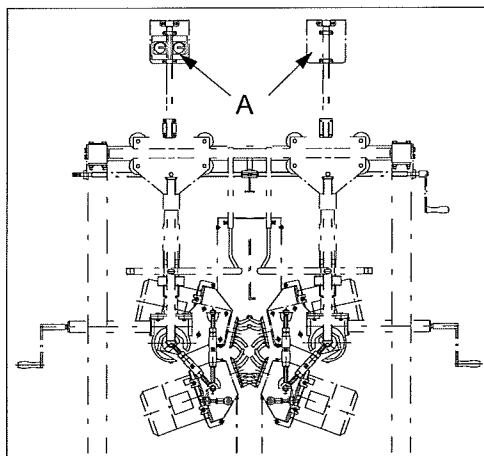


Figure 3.2 The main switch

Electrical connection

The main switches (A) for the motors are situated on the top of the plucker. The motor wiring is led safely upwards through special sleeves.

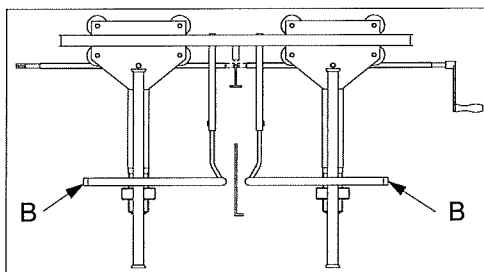


Figure 3.3 Water pipes.

Water connection

The water pipe connections are situated on the side of the water pipes. Attach a hose to the water supply tap. Attach the other end of the hose connector (B) using a hose clamp. Turn on the water tap.

3.3 Sense of rotation of the plucking discs.

The plucking discs must be driven in opposite directions of each other! See also figure 3.2. In case one plucking beams is divided into two drive belts, each driven by a motor, the two central adjacent discs must also driven in opposite direction of each other! If necessary change the polarity of a motor.

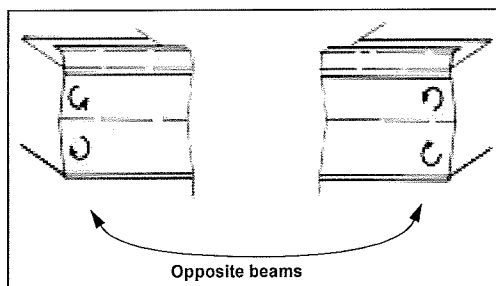


Figure 3.4 Rotation of the plucking discs.

- Sense of rotation of the plucking discs.

4 Operation

This chapter gives a description of the actions that must be taken before operation.

4.1 The manual controls

1. On/off switch.
2. Emergency stop switch. This is a function for the whole line. See also the manual "Overhead conveyor" section 5.2.
3. Cleaning switch. This is a function for the whole line.

4.1.1 Functions of the manual controls

The list below gives a short description about the functions of the manual controls. The sequence corresponds to the numbering used in section 4.1.

1. If you put the switch on, a green lamp is on. The machine is working now.
2. In case of an emergency you must use the emergency stop switch. The emergency stop switch stops the whole line. The machine will stay in the last position. A central red lamp is on.
3. You must operate this switch before cleaning the machine.

4.2 The start-up mode

The machine is part of the conveyor system. This is controlled from the main switch box. Here you can find the start/stop switch of the machine.

The switch is a selection switch; 1= Start, 0 = Stop. After selecting "1" the machine starts running and is ready for production.

4.3 In production

When the machine is correctly adjusted it will continue to function independently. However, it is important to follow the cleaning procedures and make any adjustments necessary for different size and weight of broilers.

4.4 The stop mode

The start/stop switches of the machine is located on the central switch box of the line. Before stopping the machine it is advised to stop the supply of birds and to transport the present birds away. Rotate the start/stop switch to "0" to stop the machine.

4.5 The emergency stopping device (if applied)

For use in emergency situations, emergency stop switches are mounted on the line. These switches stop immediately the whole line within the conveyor track and all the machines. The emergency stop switches may only be used in case of an emergency. See also the manual "Overhead conveyor" section 5.2. for more information

For re-starting the machine after using an emergency stop switch you must first reset the switch. The procedure is as follows:

- Cancel the emergency situation.
- Reset the emergency stop switch.
- Reset the emergency stop switch circuit by means of the blue button on the panel.
- Accept the fault on the touch screen.
- After this you can restart the machine.

5 Cleaning

This chapter gives a description of the cleaning activities that must be carried out to keep the machine in good condition and to facilitate safe and hygienic operation.

5.1 Cleaning preparations

- During cleaning you are advised to wear detergent- and disinfectant proof clothing. Protect your eyes and skin.
- Ventilate the room in which you are working
- After production, all products should be removed from the room in which the machine is installed. Remove coarse dirt.



Switch off the main switch of the machine before cleaning/adjustments.

5.1.1 Disassembly/assembly for cleaning

To promote hygiene, the plucker has a hanging frame with tilting plucking boxes. A plucking box can be tilted through 180° using a winch. This makes it easy to clean the inside of the plucker.

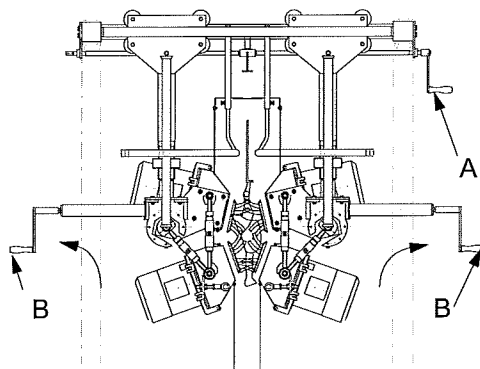


Figure 5.1 Tilting plucking boxes.

- Rotate crank (A) so that the distance between plucking units is reached.
- Rotate crank (B) so that the plucking box will be tilted through 180°.
- The inside of the plucker can now be cleaned.
- After cleaning, these steps are reversed to reassemble.

5.2 General cleaning schedule

<i>During breaks</i>	<i>Daily after production</i>	<i>Monthly *</i>
High-pressure water cleaning.	High-pressure water cleaning. Alkaline foam cleaning. High-pressure water cleaning. Desinfection. High-pressure water cleaning.	Acid foam cleaning.

Table 5.1 The cleaning schedule of the machine.

*) or as often as necessary.

- Clean the machine from top to bottom.
- During cleaning, pay extra attention to:
 - Plucking discs.
- During cleaning, never point the high-pressure water jet to the control panel or motor.
- For proper functioning, after cleaning the moving parts of the machine should be sprayed with a vegetable oil.

5.2.1 High-pressure water cleaning

With high-pressure water cleaning nearly all visible dirt will be sprayed off the machine. This cleaning is a preparation for the daily foam cleaning and disinfection. Furthermore, a high-pressure water jet can be used for localized cleaning during production breaks.

- High-pressure water cleaning should always be performed as close to the machine as possible. Stop the machine before starting the high-pressure water cleaning
- Prevent people from getting in the way of the high-pressure jet. Always hold the spray gun with both hands.

For high-pressure cleaning use the following equipment:

- A fixed piping system, delivering water-pressure with a minimum of 3,000 kPa and a maximum of 6,000 kPa and with a water temperature of approx. 55°C. at the spray-mouth
- A spray lance with a downward bent nozzle. Such a spray lance has the advantage of not transporting the dirt upwards
- A spray nozzle of the flat-jet type with a spray-angle of 15° to 25° and a water consumption of 800 to 1000 litres/hour at a water pressure of 6,000 kPa

A water temperature of 55°C is advised, since a higher temperature may cause sticking of proteins and with a lower temperature the fat will not be removed properly. The pressure should not be too high, to avoid unnecessary mist, and to protect e.g. bearings.

The unit kPa (kilopascal) is SI standard for pressure:

<i>New</i>	<i>Old</i>
100 kPa	1 bar
3000 kPa	30 bar
10000 kPa	100 bar

Table 5.2 **SI.**

5.2.2 Alkaline foam cleaning

Alkaline foam cleaning ensures that fat, proteins etc. will be degraded chemically.

- Foam the machine from top to bottom. Foaming may be performed while the machine is running slowly, provided sufficient distance is kept to the moving parts of the machine. Mind the other machines.
- After foam cleaning remove the dirt and foam residue thoroughly by means of the high-pressure waterjet.

For alkaline foam cleaning use the following equipment:

- A fixed piping system for foam delivery.
- A large spray-nozzle of the flat-jet type, through which the foam can be distributed equally along the surface.
- The alkaline foam detergent used should not affect the materials which it will contact. Consult your detergent supplier to know which detergents will fulfil these specific requirements and which cleaning directions to follow. Obey these instructions. Incorrect application may be dangerous

Please avoid:

- A too high concentration.
- A too long contact
- Not rinsing
- Mixing with other products

It is advisable not to use a combination of detergents and disinfectants ('detergent sanitizers'). This leaves the machine too dirty to be disinfected effectively.

A combination of these two methods is of no use at all.

5.2.3 Disinfection

Alkaline foam cleaning alone does not lead to a complete killing and removal of bacteria, fungi, viruses, algae and yeasts. For that purpose a separate disinfectant is needed. Disinfect the machine daily, after the alkaline foam cleaning.

- Disinfection only makes sense when the surfaces are visibly clean after the alkaline foam cleaning.
- Spray the disinfectant onto the machine and surrounding area. When disinfecting, it is difficult to judge just how thorough you have been.
- We advise you to disinfect the machine and the area immediately surrounding it in an established pattern.
- After applying, the remaining foam rests should be removed by means of high-pressure water jet.



Damage may be caused by not thoroughly rinsing the disinfectant. Evaporation of chlorous drops will lead to a very high concentration of chlorine. In combination with long contact-times can cause corrosion to metals.

For disinfection use the following equipment:

- A fixed piping system.
- A spray lance with downward-bent top
- A spray nozzle of the flat-jet type, with a spray angle of 15° to 25°
- The used disinfectant should not affect the materials which it will contact. Consult your detergent supplier to know which detergents will fulfill these specific requirements and which cleaning directions to follow. Obey these instructions. Incorrect application may be dangerous.

5.2.4 Acid foam cleaning

"Hard" water causes calcium and rust deposits on the machine. In order to remove these, a regularly performed acid foam cleaning is necessary. Depending on the hardness of the water this may be either once a month or more often if necessary.

- Foam the machine from top to bottom. Foaming may be performed while the machine is running, provided sufficient distance is kept from the moving parts of the machine. Mind the other machines and the overhead conveyor.
- After affecting, the dirt and foam residue should be removed thoroughly by means of the high-pressure water jet.



Contact between acids and chlorous products results in a poisonous gas. Ensure that these products do not come in contact with each other. Ventilate the room in which you are working.

For acid foam cleaning use the following equipment:

- A fixed piping system for foam supplying.
- A large spray-nozzle of the flat-jet type, that distributes the foam equally along the surface.
- The used acid foam detergent should not affect the materials with which it comes into contact. Consult your detergent supplier which detergents fulfil these specific conditions and the concerning directions for operation. Obey these instructions carefully. An incorrect application may be dangerous.

It is advisable not to use springwater for the production of foam; this water may contain a high degree of chloride. Tap water reduces the chance of rust.

5.3 Advising cleaning chemicals to the customer.

Customers sometimes ask Meyn Food Processing Technology B.V. to advise them about cleaning chemicals and disinfection chemicals. As a result of these questions some recommendations are as follows as stated below.

We recommend:

- To contact the major manufacturers or suppliers of these chemicals, like Henkel, Diversey etc.. These companies have a good knowledge of cleaning chemicals and their corrosive or destructive properties on materials.
- Do not buy unknown (cheap) products.
- To follow all user instructions carefully.
- Meyn Food Processing Technology does not recommend to use specific cleaning chemicals, because we do not know which chemicals are available in each specific country. We can not know the (secret, complex) composition of the products. Besides this, Meyn Food Processing Technology does not accept any responsibility for any damage caused by wrong application of cleaning chemicals.
- The above mentioned can only be avoided by a direct contact between user and supplier of cleaning chemicals.

6 Maintenance and trouble shooting table

This chapter describes the maintenance activities that are necessary to keep the machine in good condition and to facilitate safe operation. We advise you to maintain the machine regularly. This lengthens the working life of the machine and reduces the chance of failures.

6.1 Lubricants

For lubrication of the points mentioned in the lubrication point list only use lubricants which are approved for the food industry.

6.2 Checklist

<i>Part</i>	<i>Check</i>
Daily, The pluck fingers	Check for wear. If necessary, replace the plucking fingers.
Weekly, the drive belts.	Check for wear and belt tension. If necessary, replace the belts.

Table 6.1 Checklist.

6.3 Lubrication-point list

<i>Part</i>	<i>lubricant</i>
Daily	None
Weekly, all adjustments spindels.	Lubricate with a little oil to ensure smooth working.
Monthly	None

Table 6.2 Lubrication-point list

6.4 Trouble shooting table

<i>Failure</i>	<i>Cause</i>	<i>Solution</i>
Machine fails to function.	• Electrical power failure.	• Check that the machine is properly connected and follow the start-up procedure.
The overhead conveyor does not work.	• Emergency stop pressed. • Motors are defective.	• Resolve the emergency situation and reset the emergency stop. • Replace the motors. See chapter 7.
Feathers are not removed.	• Scalding Temperature too low.	• Increase scalding temperature.
Epidermis not removed (at high scalding temperature.)	• Plucking line capacity too low. • Too little air supplied to the scalding. • Too little spray water during the process. • Plucker not properly adjusted. • Plucking fingers too soft. • Plucking fingers worn.	• Reduce the line speed. • Clean the air pipes. • Clean the water pipes. • Adjust the plucker correctly. See chapter 7. • Use a harder type of plucking finger. • Replace the plucking fingers. See chapter 7.

Failure	Cause	Solution
Skin of chicken damaged.	• Plucking too aggressive.	• Increase the distance between the plucking units. See chapter 7. • Use softer plucking fingers.
White patches (burn areas).	• Scalding temperature too high. • Insufficient spray water during the process.	• Reduce scalding temperature. • Clean the water pipes. Increase the water supply.
Broken wings.	• Plucking too aggressive. • Incorrect plucker adjustment.	• Reduce the speed of the plucking disk. • Use softer plucking fingers. • Adjust the plucker correctly. See chapter 7. • Use more pluckers. Smaller plucking disks can then be used, which process the chickens less aggressively.
Broken legs	• Birds swinging excessively in the plucker	• Adjust the water pipes/guides correctly so that the birds do not swing too much. see chapter 7.
Feathers are not removed from the legs.	• Plucker not correctly adjusted. • Legs are not properly scalded.	• Adjust the plucker correctly. See chapter 7. • Raise the water level in the scalding tank.
Heads removed from the chickens.	• The killer cut is too deep.	• Adjust the killer cut to be less deep.
One leg not on the shackle.	• The sense of direction of the discs is wrong. • Water pipes/guides are set too high.	• Check the sense of direction of the plucking discs. See chapter 3. If necessary change the polarity of a motor. • Set the water pipes/guides lower. See chapter 7.

<i>Failure</i>	<i>Cause</i>	<i>Solution</i>
Too many birds pushed out of the shackles accidentally during plucking.	• The sense of direction of the discs is wrong.	• Check the sense of direction of the plucking discs. See chapter 3. If necessary change the polarity of a motor.

Table 6.3 Trouble shooting table.

7 Disassembly/assembly and adjustments

This chapter will describe how to disassemble, assemble and adjust components of the machine after maintenance and/or replacement of parts.

7.1 Basic adjustments

The basic adjustments of the various parts of the units in view of each other are already done in the factory. However, as a result of the changing of certain parts, c.q. local circumstances, it may be necessary to re-insert, c.q. readjust these basic adjustments.



Disconnect the motors from the mains before cleaning/adjusting.



Place warning signs with the control panels of the machine and the overhead conveyor.



Basic settings of the machine, performed by MEYN are for birds weighing (alive) approx. $\pm 2,000$ gram.

Tip!

Put marks on the parts to be readjusted before changing the basic settings. This will save you a lot of time and annoyance.

7.2 Adjustments with machine not running

7.2.1 Height adjustment

The height to which the plucker must be adjusted depends on the height at which the chickens enter the machine. Depending on the desired result, you can set the ideal height yourself. To achieve this, it is conceivable that the plucker might be installed at an angle.

The following can be taken as a reference height. Take the plucker finger that is closest to the shackle and bend it towards the shackle. If the plucker finger nearly touches the shackle, the height is adjusted correctly.

The height can be set in three positions. If the desired height cannot be achieved with the current setting, the height of the machine can be adjusted.

Height adjustment is achieved by means of a movable tube 'tilt plate (A)' with three holes at different heights. The tube slides in a second tube adjustment frame (B), which has a slotted hole. When the desired height has been determined, the tube is secured with a shaft (C). A special adjustment nut (D) can then be used to make fine height adjustments.

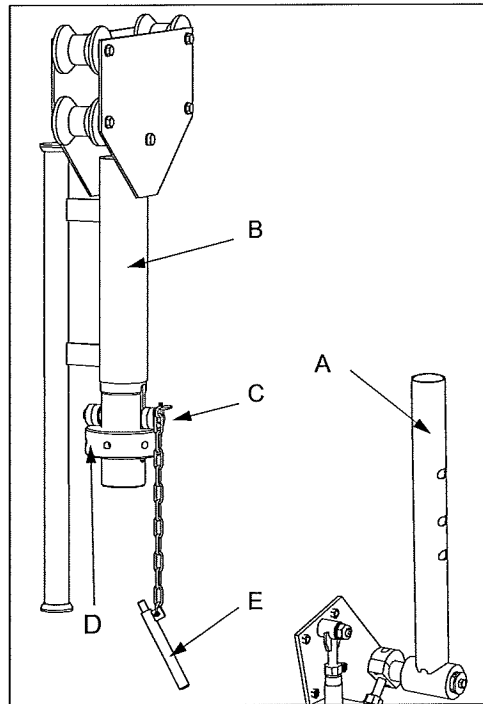


Figure 7.1 Height adjustment.

Height adjustment

- Withdraw the securing shaft (C).
- Move the plucking unit to the desired height.
- Re-insert the securing shaft (C).

Fine adjustment

- Insert the pin (E) into the nut (D).
- Rotate the nut (D)
- Depending on the direction of rotation, height can be increased or decreased by a small amount.

7.2.2 Width adjustment

The width between the plucker cabinets to which the plucker must be adjusted depends on the type of chicken to be plucked. Depending on the desired result, you can set the ideal width yourself. To achieve this, it is conceivable that the plucker might be installed at an angle.

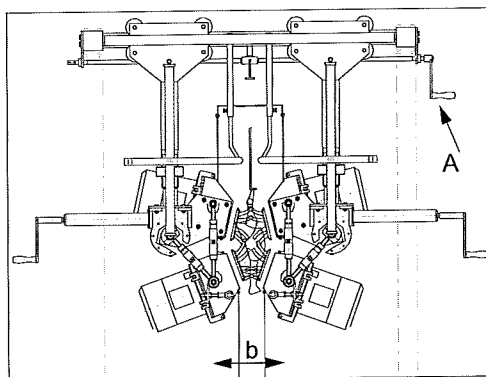


Figure 7.2 Crank handles

- Rotate the crank handle (A).
- Depending on the direction of rotation, the distance (b) between the plucking cabinets will increase or decrease.

Attention: The fingers may neither touch each other, nor the shackles.

7.2.3 Angular adjustment

The angle to which the plucking beams must be adjusted depends on the type of chicken to be plucked. The angle between the plucking beams can be changed by rotating the tensioners (A and B). Rotating the tensioner (B) will also change the distance between the beams. To give a reference value, do the following: imagine the plucking beams are initially positioned vertically above one another; the lower beam must then rotated by 30° from this initial position.

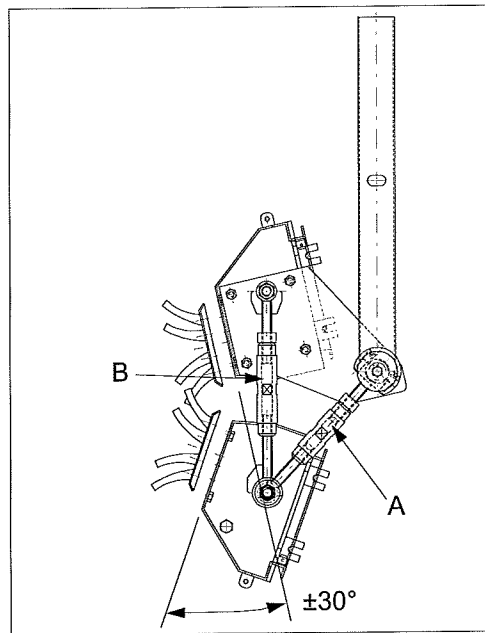


Figure 7.3 Angle adjustment.

Angular adjustment

The 30° angle is a reference value. You can adjust the angle by trial and error to suit your own requirements.

- Rotate the tensioners A and B.
- You can change the angle to suit your own requirements.

7.2.4 Water pipes

The water pipes have various functions. They ensure that the plucking fingers are kept continually wet, which prevents them from drying out. The water pipes also ensure that the feathers are continually flushed away. In addition, the water pipes serve as guides, so that the chickens hang in a more stable manner in the plucker.

It is important to clean the water pipes regularly, so that a continuous supply of water is assured. If there is a change in the average weight of the chickens to be plucked, then the water pipes, which act as guides, must be adjusted. The reference values are shown in Figure 7.7 below. Using these initial values, you can adjust the machine by trial and error to suit your own conditions.

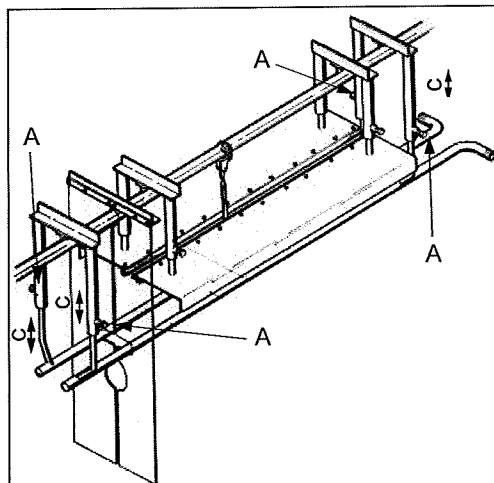


Figure 7.4 water pipes.

The height of the water pipes.

When the cap screw (A) is loosened, the water pipe can be moved vertically.

- Loosen the cap screw (A).
- Slide the water pipe to the desired position (c).
- Tighten the cap screw (A).

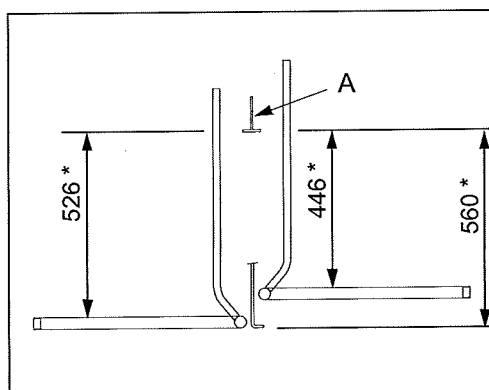


Figure 7.5 Reference values.

Reference values

The height of the water pipes in relation to the T-beam (A).

- The distance between the water pipes is fixed.
- The values shown in the illustration are reference values only.
- If the foot of the shackle points to the left, then the left-hand water pipe must be adjusted to be highest.

*These values are considered as the standard killing shackle is in use.

7.2.5 Plucking fingers

Plucking fingers are rubber pegs that are mounted on the plucking disks. The plucking disks are driven by the motors via driving belts. During the process, the plucking fingers follow the shape of the chicken and 'twist' the feathers out of the bird. In course of time the fingers are worn out, so you have to replace the fingers.

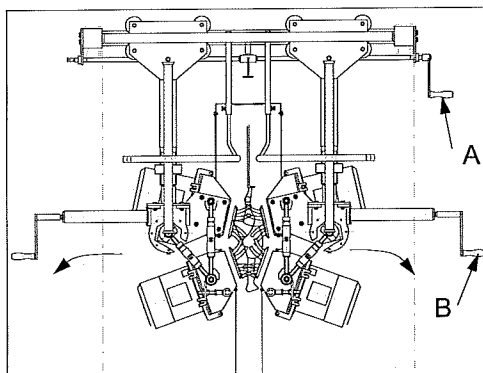


Figure 7.6

Replace a plucking finger

- Rotate crank (A) so the distance between the plucking beams increases.
- Stop rotating as soon as the extreme position is reached.
- Rotate crank (B) until the plucking box is tilted through 180°.

- Cut away the worn fingers.
- Press out the remaining material manually.
- Push the new plucking finger through the hole on the underside of the plucking disk.

Using the special plucking finger puller, pull the plucker finger pneumatically through the hole.

- Place the plucking finger puller on the plucking finger.
- Use the puller to pull the plucking finger through the hole.
- When the groove in the plucking finger is in position, the plucking finger is correctly positioned.
- If all worn fingers are changed. Rotate the plucking beams by means of the crank handles into the working condition.

7.2.6 Electric motors

The electric motors drive the plucking disks via pulleys. The motors are mounted on the motor plate with four bolts.

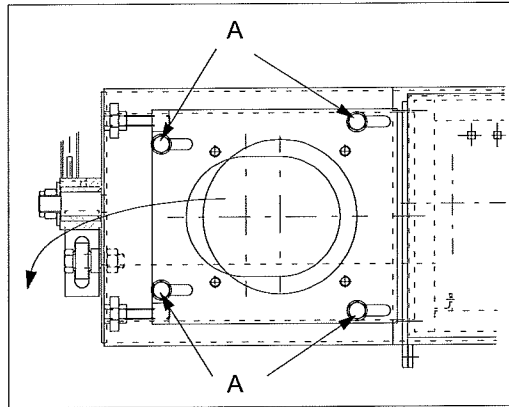


Figure 7.7

- Unscrew the four bolts (A).
- Remove the motor.
- Fit a motor in the reverse order.

7.2.7 Drive belts

The drive belts transfer power and drive the plucking disks. It is important for the belts to be correctly tensioned.

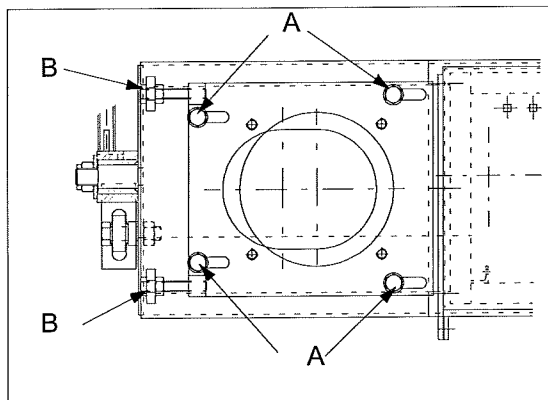


Figure 7.8

Replacing a belt

- Open the door of the plucking beams and secure it.
- Slightly loosen the nuts (A) to allow the motor to be moved.
- Slightly loosen the tensioning bolts (B).
- The belt can now be replaced.
- Re-assemble in reverse order.

When changing the drive belts you should check, before placing the new belts into the machine, whether all plucking discs are able to run freely. Malfunctioning bearings have to be changed.

Tensioning the belt:

After the belt has been replaced, it must be adjusted. the belt needs no pre-tensioning. Belt tension is adjusted by listening to the sound the belt makes. When the belt is correctly tensioned you will hear nothing. The belt is tensioned by screwing in the tensioning bolts (B).

- Switch on the motor. You will probably hear a humming sound. if there is no sound, you do not need to tension the belt.
- To tension, screw the tensioning bolts (B) in.
- When you can no longer hear any sound, the belt is correctly adjusted.
- Tighten the lock nuts on the tensioning bolts.

Appendix A: Drawings and part list

In this appendix the most important parts of the Meyn machine are described. Parts can be ordered by specifying both part number and serial number. The serial number is punched into the machine identification plate, which also shows the year of construction.

In order to facilitate the ordering of spare parts, a special spare part order sheet has been added at the end of this manual. Please use this order sheet or a copy when ordering spare parts, not only with written orders but also when ordering by telephone.

Address your orders to:

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P.O. box 16
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The Netherlands

Telephone: (+31)(0)75 684 3355
Fax: (+31)(0)75 684 9651
E-mail: spares@meyn.net
Website: www.meyn.nl